

Disorder in Cavity-Coupled Jahn–Teller Active Molecules

From single-molecule polaritons to disorder-averaged collective spectra

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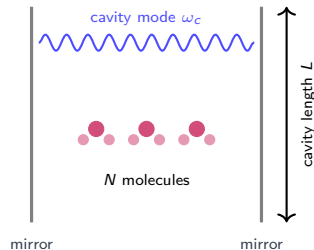
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Outline

- 1 Cavity–matter interaction
- 2 Jahn–Teller active molecules
- 3 Single molecule: cavity Jahn–Teller polaritons (2023)
- 4 Collective vibronic cascade (2025)
- 5 Introducing disorder
- 6 Disorder in our model & results

Molecules in an optical cavity: polaritons

- A Fabry–Pérot cavity confines the electromagnetic field into discrete **standing-wave modes** of frequency ω_c set by the mirror spacing L .
- When the light–matter coupling exceeds all loss rates (*strong coupling*), molecule and photon form hybrid eigenstates: **polaritons**.
- These split into a **lower** (LP) and **upper** (UP) polariton, separated by the **Rabi splitting** Ω .
- Coupling N molecules collectively enhances the interaction as $\sqrt{N}$: the single-molecule coupling is $g = \Omega/(2\sqrt{N})$.



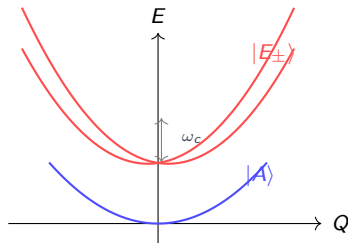
Collective strong coupling:
LP/UP split by $\Omega = 2\sqrt{N}g$.

The Jahn–Teller effect and the $(E \times e)$ model

Jahn–Teller theorem. A non-linear molecule in an orbitally degenerate electronic state spontaneously distorts to lift the degeneracy.

$(E \times e)$ problem. A doubly-degenerate electronic state E (components $|E_x\rangle, |E_y\rangle$) coupled to a doubly-degenerate vibrational mode e (coordinates Q_x, Q_y):

- ground state $|A\rangle$, excited doublet $|E_{\pm}\rangle = (|E_x\rangle \pm i|E_y\rangle)/\sqrt{2}$;
- potential surfaces form a *Mexican hat* with a **conical intersection**;
- vibronic coupling κ mixes electronic and nuclear motion $\Rightarrow$ **vibronic angular momentum** is exchanged.



Excited $(E \times e)$ surfaces (red) and ground $|A\rangle$ (blue) vs. mode coordinate Q .

Cavity Jahn–Teller Hamiltonian — one molecule

The molecule couples to two perpendicular normal-incidence cavity modes, recast as circular modes $\hat{a}_{\pm}$ carrying photon angular momentum $\pm\hbar$. The cavity–JT Hamiltonian is $\hat{H} = \hat{H}_{\text{JT}} + \hat{H}_C + \hat{H}_{CM}$, with

$$\hat{H}_{\text{JT}} = \sum_{r=+,-} [\hbar\omega \hat{b}_r^\dagger \hat{b}_r + \epsilon |E_r\rangle\langle E_r|] + \kappa [(\hat{b}_+^\dagger + \hat{b}_-) |E_-\rangle\langle E_+| + \text{h.c.}],$$

$$\hat{H}_C = \hbar\omega_c (\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-), \quad \hat{H}_{CM} = \frac{\Omega}{2} (\hat{a}_+^\dagger |A\rangle\langle E_+| + \hat{a}_-^\dagger |A\rangle\langle E_-| + \text{h.c.}).$$

Operators. $\hat{b}_{\pm} = (\hat{b}_x \mp i\hat{b}_y)/\sqrt{2}$ — ladder operators of the degenerate e vibration carrying $\pm\hbar$ vibrational angular momentum; $\hat{a}_{\pm}$ — annihilation operators of the two circularly-polarized ($\pm\hbar$) cavity photons.

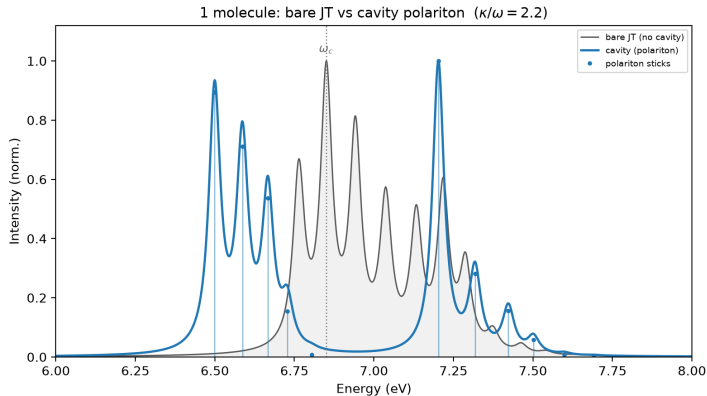
The JT (vibronic) coupling κ exchanges angular momentum *between the electronic and vibrational subsystems of the molecule*; the cavity coupling $\hat{H}_{CM}$ then exchanges it with the photon.

Angular momenta & conservation

$\hat{L}_z = \hbar(\hat{b}_+^\dagger \hat{b}_+ - \hat{b}_-^\dagger \hat{b}_-)$	vibrational
$\hat{S}_z = \hbar(E_+\rangle\langle E_+ - E_-\rangle\langle E_-)$	electronic
$\hat{l}_z = \hbar(\hat{a}_+^\dagger \hat{a}_+ - \hat{a}_-^\dagger \hat{a}_-)$	photonic

$$\boxed{\hat{J} = 2\hat{L}_z + \hat{S}_z + \hat{l}_z} \text{ is conserved.}$$

Single-molecule polaritonic spectrum (reproduced)



- Cavity ($\omega_c = 6.85$ eV) resonant with the brightest bare JT transition; the bare JT spectrum (grey) is reshaped into polaritons.
- Splits into **LP** and **UP** branches of **polaritonic peaks** — each a mixed light–matter (vibronic + photon) state, shown as sticks.
- For $N = 1$ only **3 vibronic sectors** $v \in \{0, -1, -2\}$ are accessible (bounded).

Parameters: $\epsilon = 7$ eV, $\omega = 0.082$ eV,
 $\kappa/\omega = 2.2$, $\Omega/2\omega_c = 0.05$; standalone
 $g = \Omega/2$.

N molecules in a common cavity

N identical JT molecules coupled to the two cavity modes ($\hat{H}_{m,k}$ is the single-molecule JT Hamiltonian of the previous slide):

$$\hat{H} = \hat{H}_c + \sum_{k=1}^N (\hat{H}_{m,k} + \hat{H}_{m-c,k}), \quad \hat{H}_c = \hbar\omega_c (\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-),$$

$$\hat{H}_{m-c,k} = \frac{\Omega}{2\sqrt{N}} (\hat{a}_+^\dagger |A\rangle \langle E_+|_k + \hat{a}_-^\dagger |A\rangle \langle E_-|_k + \text{h.c.}).$$

The $\sqrt{N}$ normalization keeps the collective coupling $g = \Omega/(2\sqrt{N})$ physical. Two conserved quantities:

Excitation number

$$\hat{N}_{\text{ex}} = \sum_k (|E_+\rangle \langle E_+|_k + |E_-\rangle \langle E_-|_k) + \hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-$$

Total angular momentum

$$\hat{J} = \sum_{k=1}^N \hat{V}_k + \hat{P}, \quad \hat{V}_k = 2\hat{L}_{z,k} + \hat{S}_{z,k}$$

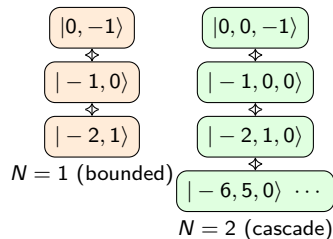
We work in the optically-relevant single-excitation sector ($n_{\text{ex}} = 1$, $j = -1$).

The collective vibronic cascade

Each molecule's JT Hamiltonian has vibronic eigenstates with a definite **vibronic angular momentum** $\nu = 2(n_+ - n_-) + S_z$:

$$\hat{H}_{m,k} = \sum_{\nu} \sum_i \lambda_i^{\nu} |\lambda_i^{\nu}\rangle \langle \lambda_i^{\nu}|_k$$

- $N = 1$: bounded — starting from the ground state only $\nu \in \{0, -1, -2\}$ are reachable.
- $N \geq 2$: the cavity photon shuttles angular momentum *between* molecules $\Rightarrow$ an unbounded ladder of ν sectors — the **collective vibronic cascade**.



The basis-state chains above show the sectors populated starting from the ground state + one RCP photon.

Participation ratio and the $P(v)$ heatmap

To *quantify* how far the cascade spreads, we decompose each polaritonic eigenstate $|\psi\rangle = \sum_{a,b,p} C_{a,b,p} |a, b, p\rangle$ over single-molecule vibronic sectors v :

Sector population $P(v)$

$$P^{(k)}(v_0) = \sum_{a: v_a=v_0} \sum_{b,p} |C_{a,b,p}|^2, \quad \sum_v P(v) = 1.$$

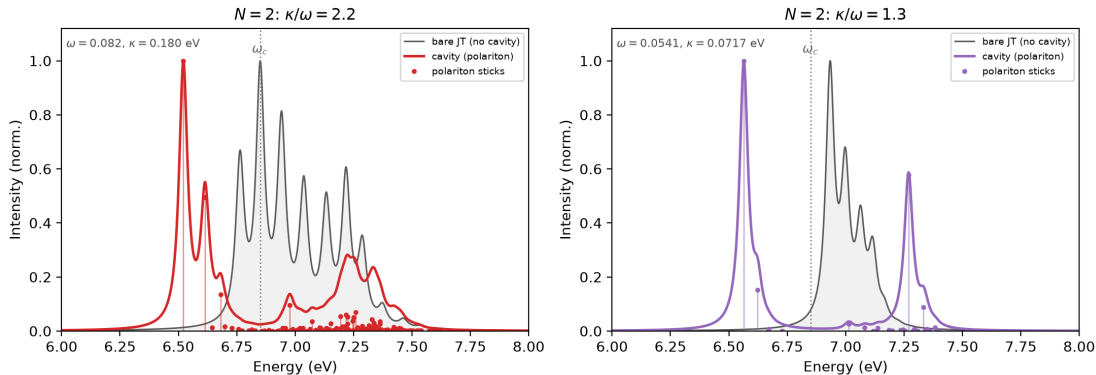
Participation ratio

$$\text{PR} = \frac{1}{\sum_v P(v)^2}$$

- PR counts how many v sectors meaningfully participate: $\text{PR} = 1$ if localized in one sector, larger if spread over many.
- $N = 1$: $\text{PR} \leq 3$ (bounded). $N = 2$: PR up to ~ 20 .
- The $P(v)$ **heatmap** plots, for every bright polaritonic state (x , energy-ordered), its $P(v)$ over sectors v (y), colored by value — the paper's Fig. S1.

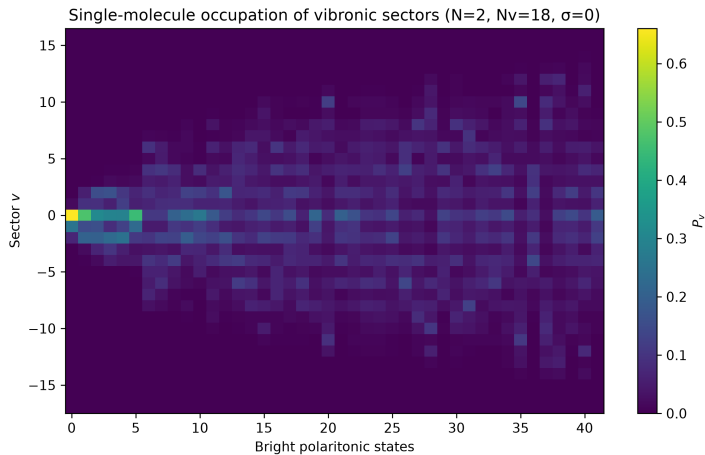
Two-molecule polaritonic spectra (reproduced)

Two-molecule cavity JT spectra (reproduced), $\omega_c = 6.85$ eV, $\Omega/2\omega_c = 0.05$



Bare JT (grey) vs. two-molecule cavity polariton spectrum (colored) with the polaritonic **stick spectrum**, for two vibronic-coupling strengths. Left: $\kappa/\omega = 2.2$ ($\omega = 0.082$, $\kappa = 0.180$ eV). Right: $\kappa/\omega = 1.3$ ($\omega = 0.0541$, $\kappa = 0.0717$ eV). Both at $\omega_c = 6.85$ eV, $\Omega/2\omega_c = 0.05$, $N = 2$.

Vibronic-sector heatmap $P(v)$ — clean $N = 2$ (reproduced)



For every *bright* polaritonic eigenstate we plot the probability $P(v)$ that a molecule occupies vibronic sector v (definition on the previous slide).

- x : bright states (energy-ordered); y : sector v ; color: $P(v)$.
- Clean $N = 2$ (Fock cutoff $N_v = 18$): a broad, well-converged cascade with $\text{PR}_{\text{max}} \approx 20$.
- This is the paper's Fig. S1 — our reference baseline.

Why disorder? Two guiding papers

Real molecular ensembles are **inhomogeneous**: each molecule has a slightly different transition energy (static *energy disorder*). Two works frame what disorder does to a cavity-coupled ensemble:

(a) Arrowhead-matrix picture (single excitation)

In the single-excitation sector the disordered Tavis–Cummings Hamiltonian is an **arrowhead matrix**: a disordered diagonal of molecular energies plus one coupling row/column to the photon. Its eigenstates are two **polariton outliers** (split by $\pm\sqrt{N}g$) plus $N - 1$ **dark states**; disorder induces *semi-localization* and reshapes the photon spectral function.

(b) Disordered Holstein–Tavis–Cummings (with vibrations)

Adds vibrations (Holstein coupling). Central result: **disorder enhances** excitation transfer and vibrational entanglement, and dark states *acquire photon weight* (brighten). Effect peaks when disorder $W \sim g_c$.

Disorder: the equations

Disordered Holstein–Tavis–Cummings model (Wellnitz *et al.*):

$$\hat{H} = \hat{H}_{\text{TC}} + \hat{H}_{\text{vib}} + \hat{H}_{\text{H}} + \hat{H}_{\text{dis}},$$

$$\hat{H}_{\text{dis}} = \sum_n \epsilon_n \hat{\sigma}_n^+ \hat{\sigma}_n^-$$

with i.i.d. energy offsets ϵ_n of mean 0 and variance W^2 (Gaussian). Consequences quantified in the papers:

- Dark states gain photon weight $\sum_{\text{dark}} |\langle d | 1_{\text{ph}} \rangle|^2 \approx W^2 / 2g_c^2 \Rightarrow$ they **appear in absorption**.
- Polaritons stay protected while g beats disorder: crossover at $W \sim 2g$.
- Arrowhead: photon spectral function $A(\epsilon) = \sum_a \text{PW}_a \delta(\epsilon - \epsilon_a)$ — this *is* the absorption spectrum.
- Eigenstate amplitudes $\psi_{a,j} \sim 1/(\epsilon_a - \omega_j)^2$; level statistics Poisson $\rightarrow$ semi-Poisson.

How disorder enters our model

We add **Gaussian energy disorder** to the two JT molecules' transition energies:

$$\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2), \quad i = 1, 2, \quad \epsilon = 7.0 \text{ eV}.$$

Per disorder *realization* we build $\hat{H}$ in the ($n_{\text{ex}} = 1, j = -1$) sector, diagonalize, take the intensity of each eigenstate as $|\langle i_0 | \psi \rangle|^2$, broaden with a Lorentzian (FWHM $\gamma = 0.025$ eV), and **average over realizations**.

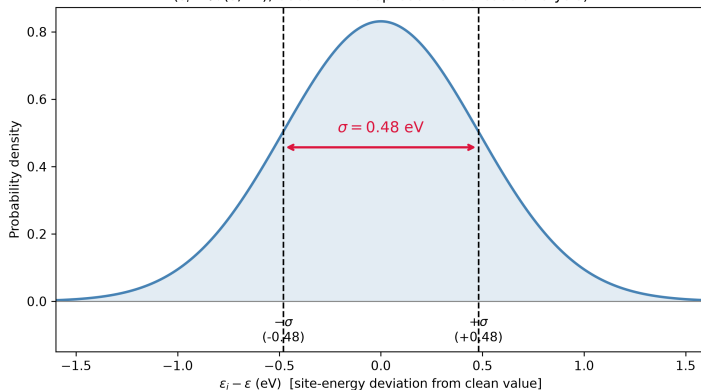
Energy scales

vibrational quantum ω	0.082 eV	JT coupling $\kappa/\omega = 2.2$
per-molecule coupling $g = \Omega/2\sqrt{N}$	0.242 eV	cavity $\omega_c = 6.85$ eV
Rabi splitting $\Omega = 2\sqrt{N}g$	0.685 eV	molecular excitation energies $\epsilon = 7.0$ eV, $N = 2$

Disorder strength quoted relative to g : we run at $\sigma \approx 2g$ (strong-disorder band), where the representative cases are physically distinct.

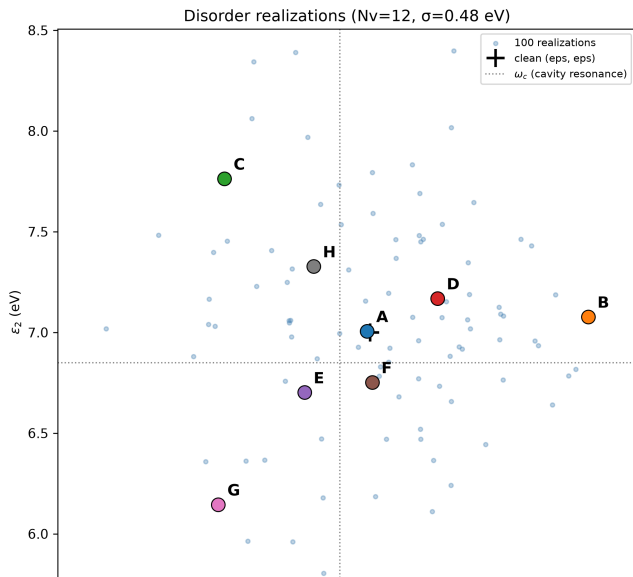
Disorder realizations: the Gaussian distribution

Gaussian disorder distribution, $\sigma = 0.48$ eV
($\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$, used in the representative-case analysis)



- Each molecule's excitation energy is drawn from $\mathcal{N}(\epsilon, \sigma^2)$ with $\sigma = 0.48$ eV.
- $\sigma/g \approx 2.0$, $\sigma/\Omega \approx 0.70$ — just below the $W = 2g$ crossover.
- Dashed lines / arrow mark $\pm\sigma$ (-0.48 to $+0.48$ eV).
- We average absorption over 100 such realizations.

Representative-realization analysis: the (ϵ_1, ϵ_2) cloud



Behind the disorder average lie many individual (ϵ_1, ϵ_2) draws. We pick **9 physically meaningful realizations** (Cases A–I), each best satisfying a criterion on the actual drawn energies:

- average shift $s = (\delta_1 + \delta_2)/2$ vs. $\sqrt{N}g$;
- inter-molecular mismatch $\Delta = \epsilon_1 - \epsilon_2$ vs. g .

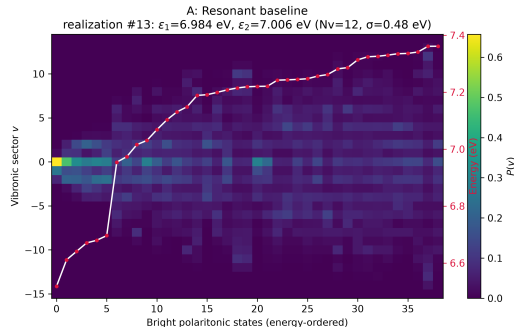
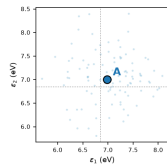
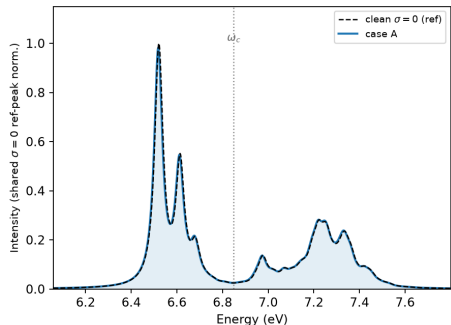
Dotted lines mark ω_c ; the “+” marks the clean (ϵ, ϵ) . Each case is then recomputed individually.

The eight representative cases ($\sigma = 0.48$ eV)

Case	Description	(ϵ_1, ϵ_2)	Δ	#bright	verdict
A	Resonant baseline (no disorder)	(6.98, 7.01)	0.02	39	$\approx$ clean
B	Large mismatch (localized)	(8.08, 7.08)	$4g$	11	dead
C	Opposite disorder	(6.28, 7.76)	$6g$	14	partial $\rightarrow$ dead
D	Common blue detuning	(7.33, 7.17)	0.17	12	partial
E	Common red (matched)	(6.67, 6.70)	0.03	104	enhanced
F	Single-molecule resonant	(7.01, 6.75)	g	132	enhanced
G	Bare-molecule limit	(6.25, 6.15)	0.10	8	dead
H	A-C line, closest to ω_c	(6.72, 7.33)	$2.5g$	46	intermediate

#bright = number of bright vibronic (polaritonic) states above 1% of the peak. Clean $N = 2$ reference: **42**. Disorder can *brighten* dark states (E, F, H $> 42/39$) or kill the cascade (B, G).

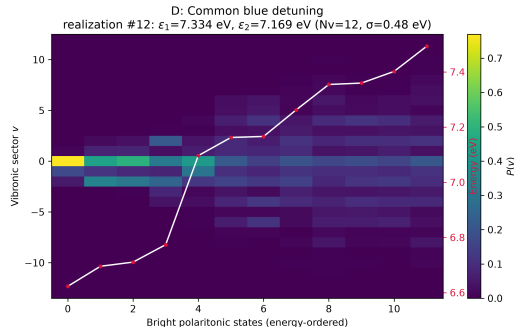
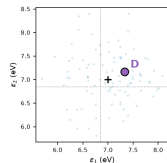
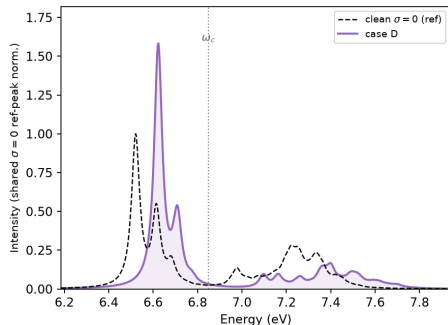
Case A — resonant baseline (no disorder)



Starting point: both molecules essentially on resonance, $\epsilon_1 \approx \epsilon_2 \approx \epsilon$.

$(\epsilon_1, \epsilon_2) = (6.98, 7.01)$, $\Delta = 0.02$, $s \approx 0$; **39 bright**, $\text{PR}_{\text{max}} = 20.6$ at $E = 7.361$ eV — reproduces the clean collective cascade.

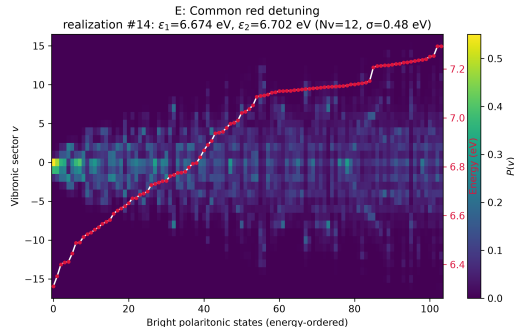
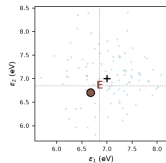
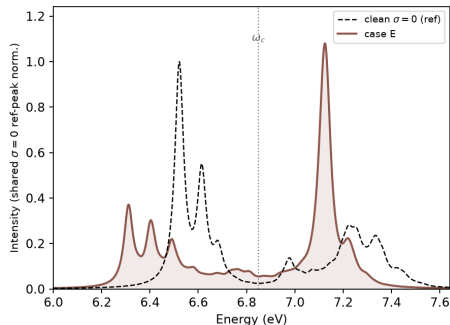
Case D — common blue detuning



From A, shift *both* molecules *up* together by $\sim +g$ (matched, Δ small): a common blue shift.

$(\epsilon_1, \epsilon_2) = (7.33, 7.17)$, $\Delta = 0.17$, $s = +1.0g$; **12 bright**, $\text{PR}_{\text{max}} = 12.5$ at $E = 7.354$ eV — **partial**: bright manifold pushed partly off the photon.

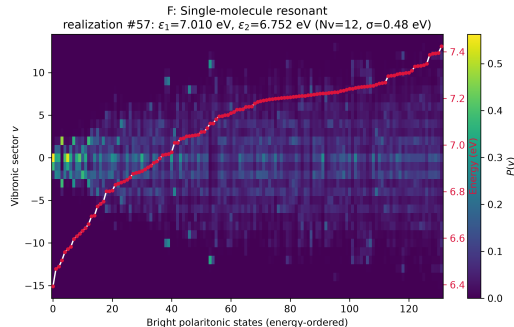
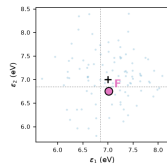
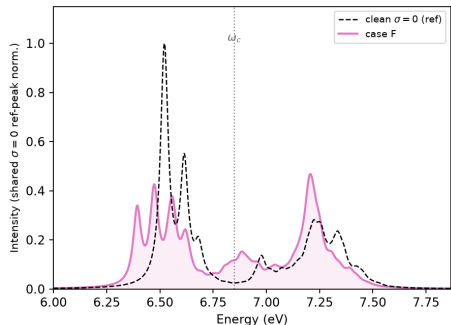
Case E — common red detuning (matched)



Flip the common shift to $-g$ (mirror of D, but red): this lands in the enhancement band, pushing the bright polariton *into* the dense dark manifold.

$(\epsilon_1, \epsilon_2) = (6.67, 6.70)$, $\Delta = 0.03$, $s = -1.3g$; **104 bright**, $\text{PR}_{\text{max}} = 22.4$ at $E = 7.215$ eV — **enhanced**: disorder brightens the whole dark band.

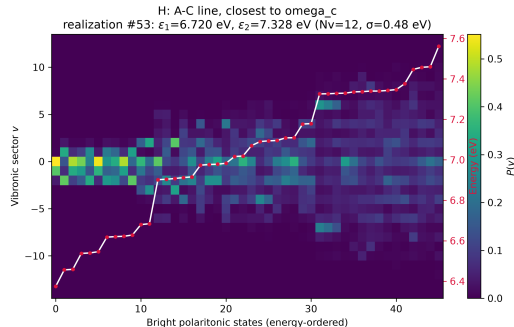
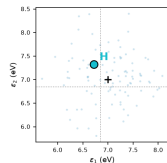
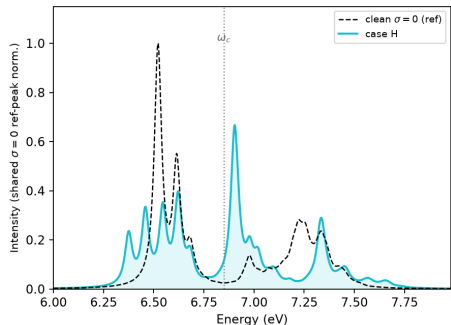
Case F — single-molecule resonant



Now introduce a real inter-molecular mismatch: bring one molecule back *onto* resonance while the other stays detuned by $\gtrsim g$ — a mildly asymmetric *collective* case (not localized).

$(\epsilon_1, \epsilon_2) = (7.01, 6.75)$, $\Delta = g$, $s = -0.5g$; **132 bright**, $\text{PR}_{\text{max}} = 21.6$ at $E = 7.424$ eV — **enhanced** (maximal dark-state brightening).

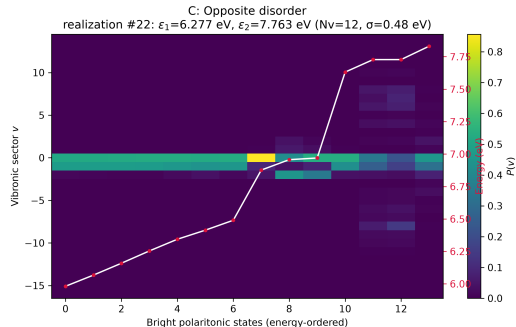
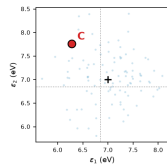
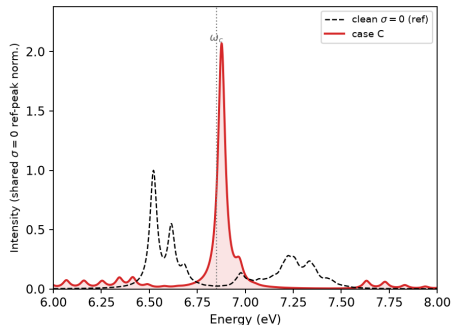
Case H — A-C line, interior (opposite disorder)



Detune the two molecules *oppositely* (one red, one blue) about resonance, $s \approx 0$ — the interior of the line joining A and C.

$(\epsilon_1, \epsilon_2) = (6.72, 7.33)$, $\Delta = 2.5g$, $s \approx 0$; **46 bright**, $\text{PR}_{\text{max}} = 16.2$ at $E = 7.337$ eV — **intermediate**: brightens more peaks than A yet spreads each less.

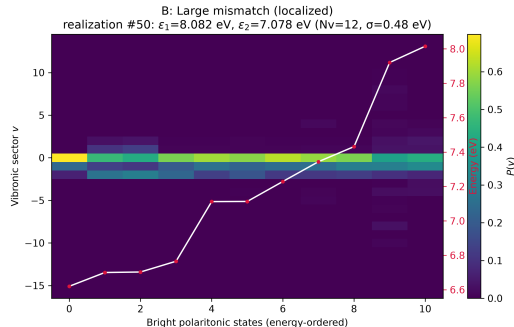
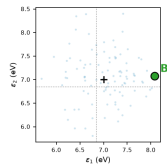
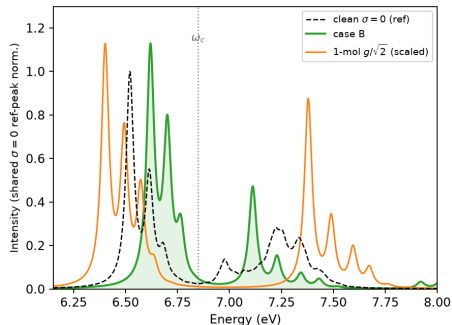
Case C — opposite disorder (far)



Push the opposite detuning *further out* along the same A–C line ($\Delta=6g$): both molecules now leave the bright line.

$(\epsilon_1, \epsilon_2) = (6.28, 7.76)$, $\Delta=6g$, $s \approx 0$; **14 bright**, $\text{PR}_{\text{max}} = 8.5$ at $E = 7.728$ eV — **partial**→**dead**: cascade collapses to a near-bare cavity peak.

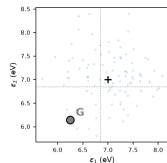
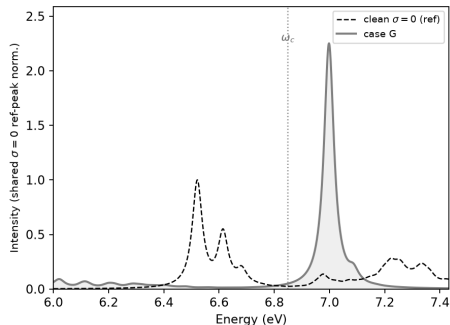
Case B — large same-side mismatch (localized)



Instead of opposite, detune one molecule far on the *same* side ($\Delta = 4g$): the excitation *localizes* on the resonant molecule $\Rightarrow$ spectrum $\approx$ a single JT-polariton (orange: 1-mol $g/\sqrt{2}$).

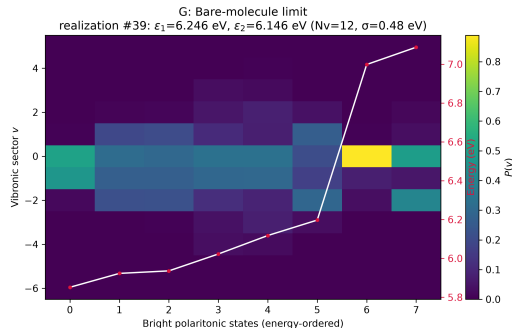
$(\epsilon_1, \epsilon_2) = (8.08, 7.08)$, $\Delta = 4g$, $s = +2.4g$; **11 bright**, $\text{PR}_{\text{max}} = 3.8$ at $E = 7.921$ eV — **dead** (single-molecule bound).

Case G — bare-molecule limit (decoupled)

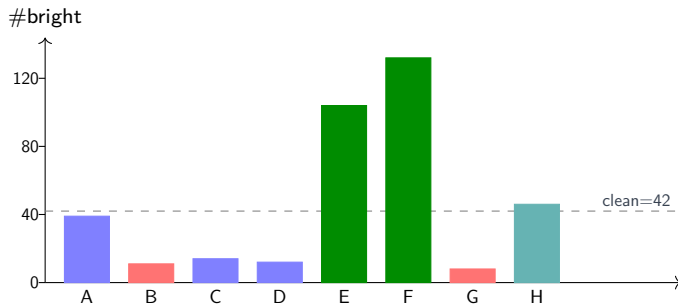


Finally push *both* molecules far below resonance together ($|s| \gg \sqrt{N}g$): both decouple from the cavity.

$(\epsilon_1, \epsilon_2) = (6.25, 6.15)$, $\Delta = 0.10$, $s = -3.3g$; **8 bright**, $\text{PR}_{\text{max}} = 4.5$ at $E = 6.118$ eV — **dead** (near-bare photon).



Number of bright vibronic states vs. case



- **Enhanced** (E, F): matched / single-resonant $\Rightarrow$ disorder brightens the whole dark manifold ($104, 132 \gg 42$).
- **Intermediate** (H): opposite disorder on the A–C line (46).
- **Partial** (A, C, D).
- **Dead** (B, G): localization / decoupling collapse the cascade (≤ 11).

Summary

- 1 **Reproduced** the single-molecule cavity-JT polariton spectrum (LP/UP, bounded ν) and the two-molecule collective spectrum + $P(\nu)$ heatmap (unbounded cascade, PR ~ 20).
- 2 **Introduced disorder** following the arrowhead-matrix and disordered Holstein–Tavis–Cummings frameworks: Gaussian excitation-energy disorder $\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$, dark states brighten, cascade protected while g beats σ .
- 3 **Implemented & analyzed** disorder in our $N = 2$ model: realizations, representative cases (A–H), and $P(\nu)$ heatmaps with an LP/UP energy overlay.
- 4 **Key result:** the number of bright vibronic states and the cascade PR track the two disorder axes — *enhanced* (E, F, H), *partial*, or *dead* (B, G).

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